24k-0918
BCS-5A
CN

ASSIGNMENT-II

TASK-1

1- Reliable data transfer is important bcz the network carries casualty lists, supply requests, and rescue assignments. If a packet is lost, corrupted or received out of order, the rescue team may receive incorrect information Therefore packets must be delivered correctly in order.

2- Stop-&-Waits sends only one packet & waits for its ACK before sending the next packet. In this scenario, the RTT is 500ms, while transmitting an 8000-bit packet at 2Mbps takes only 4ms. Therefore, the sender spends most of its time waiting instead of transmitting data.

3- Pipelining allows the sender to transmit multiple packets before receiving their ACKs. This reduces waiting time & improves link utilization. It is useful on this satellite link bcz the RTT is much larger than the packet transmission time.

4- Selective Repeat (SR) is more suitable for this scenario for two reasons. First, it retransmits only the lost or corrupted packets instead of retransmitting all outstanding packets. Second, its receiver can buffer out-of-order packets & deliver them in order when the missing packets arrive, which reduces unnecessary retransmissions on the limited-bandwidth satellite link.

CamScanner

24K-0918

TASK-2

Go-Back-N (GBN):

Event	Sender	Packet	Receiver	ACK	Sender Window
1	Send Pkt 0	0	Accept pkt 0	ACK 0	[0-3]
2	Send Pkt 1	1	Accept pkt 1	ACK 1	[0-3]
3	Send Pkt 2	2	Packet loss	-	[0-3]
4	Send Pkt 3	3	Discard: out of order	ACK 1	[0-3]
5	Receive ACK 0, send 4	4	Discard if it arrives before pkt 2	ACK 1	[1-4]
6	Rec ACK 1; send Pkt 5	5	Discard if it arrive before pkt 2	ACK 1	[2-5]
7	Time out for pkt 2	2	—	-	[2-5]
8	Retransmit pkt 2	2	Accept pkt 2	ACK 2	[2-5]
9	Retransmit Pkt 3	3	Accept pkt 3	ACK 3	[3-6]
10	Retransmit pkt 4	4	Accept pkt 4	ACK 4	[4-7]
11	Retransmit pkt 5	5	Accept pkt 5	ACK 5	[5-8]
12	Send remaining pkt as window advances	6-9	Accept in order	ACK 6-9	advances until complete

Selective Repeat (SR):

Event	Sender	Packet	Receiver	ACK	Sender Window
1	Send pkt 0	0	Accept & deliver pkt 0	ACK 0	[0-3]
2	Send pkt 1	1	Accept & deliver pkt 1	ACK 1	[0-3]
3	Send pkt 2	2	Packet lost	-	[0-3]
4	Send pkt 3	3	Buffer pkt 3	ACK 3	[0-3]
5	Rec ACK 0, send pkt 4	4	Buffer pkt 4 if pkt 2 still missing	ACK 4	[1-4]
6	Rec ACK 1, send pkt 5	5	Buffer pkt 5 if within rec window	ACK 5	[2-5]
7	Timeout for pkt 2	2	—	-	[2-5]
8	Retransmit pkt 2 only	2	Accept pkt 2; deliver buffered packets in order	ACK 2	ACKs are processed
9	Process ACKs for buffered pkts	3-5	Deliver any remaining buffered pkts	individual ACKs	window advances
10	Send remaining pkts as window advances	6-9	Accept & deliver in order	ACK 6-9	advances until complete

CamScanner

24k-0918

Comparision:

GBN: It retransmit pkt 2 & all other outstanding pkts in the sender window that have not been cumulatively ack. In the trace above, pkt 3, 4, & 5 includes.

SR: It retransmits only pkt 2, assuming the other pkts & their ACKs correct

Why Diff: GBN discards out-of-order packets, while SR buffers them & acknowledges them individually.

TASK-5

1- Packet Size, $L = 8000$ bits

Link rate, $R = 2$ Mbps $= 2000000$ bits/s

RTT $= 500$ ms $= 0.5$ s

Window size, $N = 4$

$$\frac{L}{R} = \frac{8000}{2000000} = 0.004 \text{ s}$$

$$\Rightarrow U = \frac{4/R}{RTT + 4/R} = \frac{0.004}{0.5 + 0.004} = \boxed{0.79\%}$$

Pipelining with $N = 4$

$$U = \min\left(1, \frac{N(4/R)}{RTT + 4/R}\right) = \frac{4(0.004)}{0.504} = \boxed{3.17\%}$$

2- a) $k = 3$ bits

$$2^k = 2^3 = \boxed{8}$$

Sequence No = 0, 1, 2, 3, 4, 5, 6, and 7

b) For selective repeat, max window size is half of seq

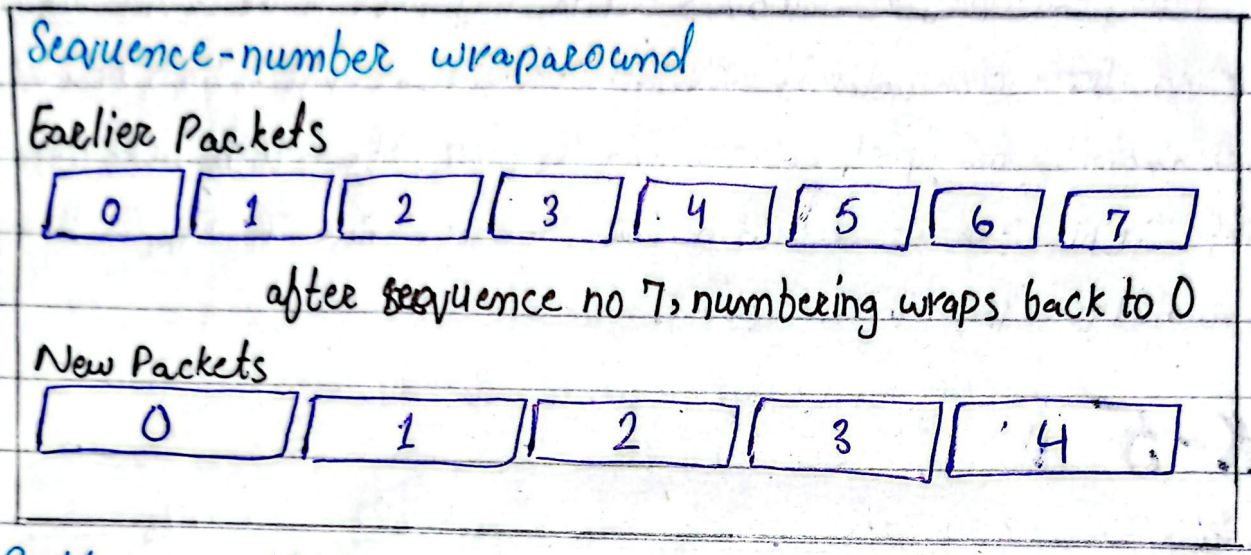
$$N_{max} = \frac{2^k}{2} = \frac{8}{2} = \boxed{4}$$

c) A larger window can overlap with sequence numbers from an ~~Either~~ earlier transmission after the sequence numbers wrap around. The receiver may mistake an old duplicate packet for a new packet with the same sequence number.

CamScanner

24K-0918

Example of an unsafe SR window size of 5



Problem: an old duplicate with sequence no 0 may be confused with a new packet numbered 0 if receiver's window overlaps.

Therefore, SR window must be no larger than 4 when using 3-bit sequence numbers.